

Network Monitoring and Health Check System

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The project was implemented in two phases:

1. Python-based prototype
2. Golang-based production-oriented versio

The Python version was developed to validate the monitoring logic and system requirements, while the Golang version focused on scalability, concurrency, reliability, and DevOps-oriented deployment

1. Python Implementation

The Python version of the project was designed as a lightweight diagnostic tool capable of performing basic system and network health checks. I use AI to accomplish this part of the project, then I check it by myself. One of the first checks performed by the application is identifying the local IP. This is achieved by creating a socket connection and allowing the operating system to determine the active network interface.

The application also retrieves the default gateway by reading the system routing. This information is useful when diagnosing network connectivity problems because it identifies the primary route used to access external networks. The application performs network connectivity tests to check the connection like TCP and public DNS servers. Calculating the available disk space on the system is also

provided and I just do it to check some code skills. I store the log of notification and status in a JSON file.

```
=====
Starting Network Health Diagnostics...
=====
Check Name           | Status   | Details
-----
Local IP             | OK       | 192.168.1.145
Gateway              | OK       | 192.168.1.1
Disk Space           | OK       | 16 GB Free
Google DNS           | FAILED   | Error: timed out
Cloudflare DNS       | FAILED   | Error: timed out
Hmrah Academy        | OK       | Connected
CPU Load             | OK       | 1.53 (1min), 1.18 (5min), 1.01 (15min)
-----
Diagnostics Completed. Report saved to health_report.json
```

2. Golang Implementation

After completing the Python implementation, I developed a more advanced monitoring system in Golang. Each configured domain is checked through both HTTP and HTTPS protocols. The application verifies the response status code and determines whether the service is operating correctly. The application performs DNS resolution for every monitored domain. This allows domain names to be translated into their associated IP addresses. Moreover, the application performs additional checks directly against the discovered IPs. This helps distinguish between DNS-related problems and service availability issues.

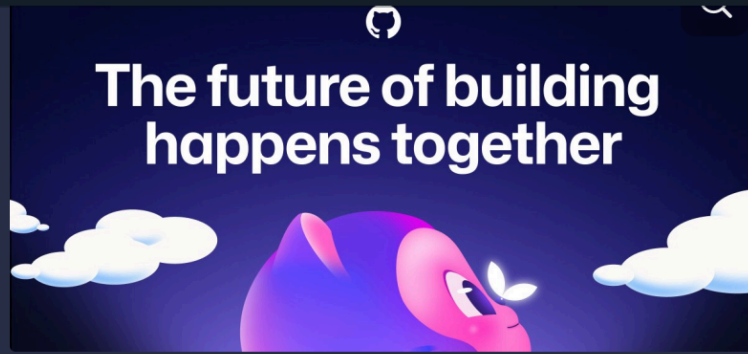
The system also performs TCP connectivity tests against critical services to verify network reachability and connectivity. We employ goroutines and synchronization mechanisms for concurrency. This significantly reduces the time required to complete monitoring cycles and demonstrates one of Go's strongest features. Failed requests are retried several times before being marked as failed. A delay is added between retries and increases after each failed attempt to reduce false alarms caused by temporary network issues. The system can also send Telegram notifications when a service remains unavailable after

all retry attempts. This helps administrators become aware of problems quickly. The monitoring service runs continuously and performs health checks at regular intervals.

```
ehsan-borhani-nia@parrot:~/Programming/Golang/practice/HamrahAcademic
File Edit View Search Terminal Help
Error: Get "http://www.google.com/": context deadline exceeded (Client.Timeout e
xceeded while awaiting headers)
2026/06/01 18:19:14 https://google.com OK
2026/06/01 18:20:01 http://github.com OK
2026/06/01 18:20:02 http://hamrah.academy OK
2026/06/01 18:20:02 https://github.com OK
2026/06/01 18:20:03 https://hamrah.academy OK
2026/06/01 18:20:09 Url: http://google.com
Error: Get "http://www.google.com/": context deadline exceeded (Client.Timeout e
xceeded while awaiting headers)
2026/06/01 18:20:40 https://google.com OK
2026/06/01 18:20:52 http://github.com OK
2026/06/01 18:20:52 https://github.com OK
2026/06/01 18:20:56 http://hamrah.academy OK
2026/06/01 18:20:58 https://hamrah.academy OK
2026/06/01 18:21:02 http://google.com OK
2026/06/01 18:21:03 https://google.com OK
2026/06/01 18:21:11 http://github.com OK
2026/06/01 18:21:11 https://github.com OK
2026/06/01 18:21:11 http://google.com OK
2026/06/01 18:21:12 http://hamrah.academy OK
2026/06/01 18:21:13 https://google.com OK
2026/06/01 18:21:14 https://hamrah.academy OK
2026/06/01 18:21:48 http://hamrah.academy OK
```

hamrahAcademy_monitoring

bot



8:39 PM



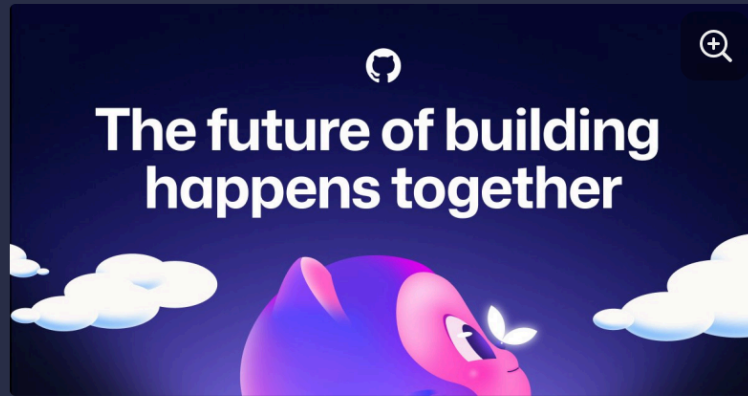
DOWN: <http://github.com>

Error: Get "https://github.com/": context deadline exceeded
(Client.Timeout exceeded while awaiting headers)

GitHub

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Join the world's most widely adopted, AI-powered developer platform where millions of developers, businesses, and the largest open source community build software that advances h...



8:40 PM



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